

## ORIGINAL PAPER

## Reinforcement Learning-Based Relay and RIS Optimization for Covert 6G Ground Station Communications

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## ABSTRACT

Dynamic channels, energy constraints, and the risk of warden detection pose significant challenges to securing covert communications in 6G ground station networks. To address these challenges, we propose a novel multi-agent reinforcement learning (MARL) framework that jointly leverages mobile relays and reconfigurable intelligent surfaces (RISs) to optimize relay trajectories, transmit power, and RIS phase shifts. Unlike traditional optimization-based and existing reinforcement learning approaches, the proposed scheme simultaneously maximizes covert throughput, minimizes energy consumption, and ensures robustness under imperfect channel state information. The optimization problem is solved using the multi-agent deep deterministic policy gradient (MADDPG) algorithm. Simulation results show that the proposed framework maintains a warden detection error probability above 90% in over 95% of time slots, while delivering covert transmission rate improvements of up to 77% compared to static-RIS baselines and 63% relative to greedy relay baselines. Moreover, the approach demonstrates resilience to channel estimation errors (e.g., the Gaussian perturbations with  $\sigma^2$  up to 0.1) and offers computational scalability, with inference time  $< 15$  ms per slot on GPU, making it suitable for real-time implementation in the next-generation satellite networks.

## 1 | Introduction

Sixth-generation (6G) wireless networks are poised to transform global connectivity through the implementation of applications such as ultra-reliable low-latency communications (URLLC), integrated sensing and communication (ISAC), and the massive Internet of Things (mIoT) [1, 2]. In this framework, obtaining global coverage requires secure ground-to-satellite links. However, due to the broadcast nature of wireless communications, these links are susceptible to interception, placing sensitive data at serious risk. Low-probability-of-detection (LPD), also known as covert communication, has emerged as a promising paradigm for concealing transmissions from adversaries [3, 4]. These covert techniques increase the likelihood that a warden will err in detection by generating signals that are

indistinguishable from background noise [5]. Nevertheless, a significant obstacle remains in the dynamic and hostile propagation environment of satellite links, which is characterized by severe path loss and time-varying fading. To overcome these challenges, solutions beyond conventional static designs are required—ones that are creative and adaptive to ensure high covert rates under stringent energy constraints and detection avoidance.

Reconfigurable intelligent surfaces (RISs) and mobile relays are two promising technologies that can enhance energy efficiency, throughput, and covertness. A RIS is a programmable meta-surface capable of altering the phase of incident signals to dynamically modify the propagation environment, thus improving reception at the intended target or suppressing signals directed at adversaries [6, 7]. Meanwhile, mobile relays, such as

unmanned aerial vehicles (UAVs), offer additional flexibility by adjusting their trajectories to maintain line-of-sight links, expand coverage, and avoid obstructions [1, 7].

Figure 1 illustrates the system model, featuring the ground station (GS), RIS, multiple UAV relays, legitimate satellite receiver, and adversarial warden.

Recent advances have highlighted the potential of RIS and reinforcement learning (RL) to enhance security for 6G networks, particularly through physical layer security (PLS) and deep RL optimization [8, 9]. For instance, a deep RL approach has been proposed to optimize secure multi-user ISAC systems using simultaneously transmitting and reflecting RIS (STAR-RIS), focusing on secure beamforming against eavesdroppers in multi-user scenarios [2]. Similarly, hybrid active-passive RIS configurations have been explored with deep RL to counter eavesdropping and jamming attacks, emphasizing adaptive reflection strategies for improved detection avoidance [4, 5].

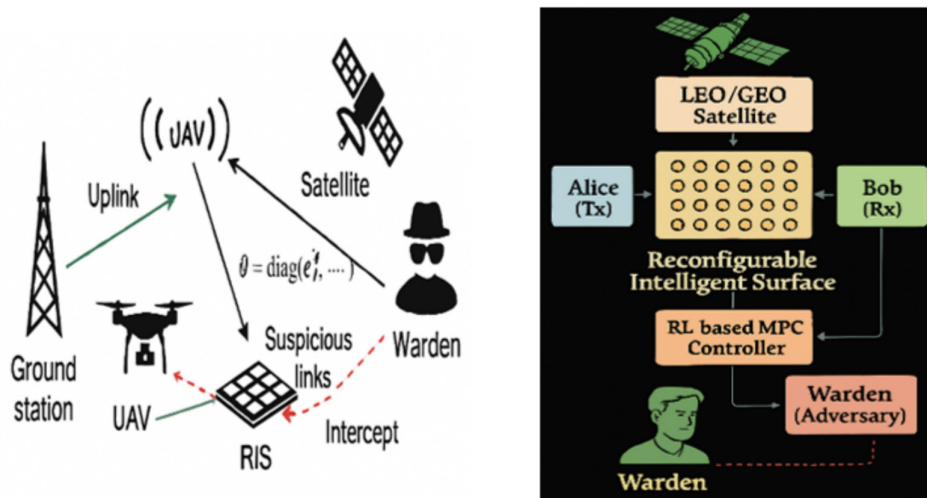
In the context of 6G IoT, deep deterministic policy gradient (DDPG)-based frameworks have integrated RIS with backscatter communication to boost secrecy rates and energy efficiency, addressing imperfect channel state information (CSI) and quantized phase shifts while adapting to dynamic channels [4]. Another study advances IoT security via RIS-enhanced backscatter, employing DDPG for dynamic beamforming that aligns reflections toward legitimate users and minimizes leakage to multi-antenna eavesdroppers, achieving higher multi-cast secrecy rates under power constraints [4, 6]. Furthermore, a DRL and phase optimization approach has been developed to secure communication networks at the physical layer, extending to multi-user and multi-eavesdropper scenarios to enhance security through adaptive phase adjustments [10].

These works underscore the efficacy of RL-driven RIS optimization for PLS in static or ground-based setups. However, they predominantly overlook the integration of mobile relays such as UAVs, joint trajectory-power-phase optimization under time-varying satellite fading, and explicit covertness constraints that balance detection error probabilities with throughput and energy use. This gap motivates our exploration of multi-agent

RL for dynamic, relay-assisted covert systems. Mobile relay-assisted adaptive connectivity and intelligent reflection based on RIS work together to create a promising paradigm for secure and efficient 6G GS networks. On the other hand, optimizing relay trajectories, transmit power, and RIS phase shifts simultaneously creates a complicated, time-coupled, and nonconvex problem. This requires maintaining a low detection probability at the warden while simultaneously optimizing the covert data rate and reducing energy consumption. This complexity is difficult to handle with traditional optimization techniques, underscoring the need for flexible and intelligent solutions. In this paper, we propose a RL framework to enhance covert communications in a 6G GS network assisted by an RIS and multiple mobile relays. We formulate the problem as a multi-objective optimization and leverage a multi-agent deep deterministic policy gradient (MADDPG) algorithm to navigate the complex solution space. Our main contributions are as follows:

- A comprehensive system model for RIS-assisted, relay-aided covert communication networks, along with a joint optimization formulation for relay trajectories, transmit power, and RIS phase shifts under dynamic covertness constraints.
- An RL-based optimization framework where the RIS controller and mobile relays act as autonomous agents in a MADDPG architecture.
- A novel covert-aware reward function that effectively balances achievable throughput, energy expenditure, and the warden's detection error probability.
- Extensive simulations demonstrating that the proposed algorithm outperforms benchmark schemes such as greedy relay selection and static RIS configurations, while achieving stable convergence.

The remaining sections of this paper are arranged as follows: The problem formulation and system model are presented in Section 2. The MADDPG-based approach suggested is described in detail in Section 3. Performance analysis and simulation results are given in Section 4. Section 5 summarizes the work and suggests areas for further research.



**FIGURE 1** | System model of RIS-assisted, mobile relay-aided covert communication in 6G ground-to-satellite networks.

## 2 | System Model and Problem Formulation

### 3 | System Architecture

The architecture of the 6G covert communication system under consideration is shown in Figure 1. A direct transmission link, relay-assisted links, and RIS-assisted reflection links are the three wireless pathways that the GS uses to connect to the satellite. To increase coverage and link reliability, mobile relays, such as UAVs or ground vehicles, adapt their trajectories on the fly. To improve signal quality and reduce detection risks, the RIS, which is made up of programmable reflecting elements, modifies the propagation environment by varying its phase shifts [8, 9]. In the meantime, a warden is supposed to keep an eye on the surroundings and use energy detection methods to spot possible transmissions. Rerouting the signal via relays and RIS increases the probability of covert transmission by masking the communication of the GS in background noise. This system model highlights the interplay between mobility, intelligent surfaces, and covertness in future 6G GS communications [11]. It is assumed that a warden is present in the environment at the same time and is equipped with energy detection techniques to detect ongoing transmissions. By rerouting and masking signals within background noise, the combination of mobile relays and RIS increases the transmission's secrecy. This system model makes it clear how mobility, intelligent surface optimization, and covertness interact. In this work, these three factors are combined to optimize the performance of covert transmission while meeting security and energy efficiency requirements.

The principal notations of the system model are summarized in Table 1, which provides a convenient reference for channel coefficients, optimization variables, and performance metrics. This concise overview complements the schematic representation in Figure 1 and helps the reader grasp the interdependencies in the effective channel formulation (cf. Equation (1)).

#### 3.1 | System Entities and Channels

#### 3.2 | System Entities

- **GS:** equipped with  $M$  antennas and a power budget  $P_{\text{GS}}$ , responsible for generating the beamforming vector  $\mathbf{w}$ .
- **Mobile Relays:**  $K$  relays with positions  $\mathbf{p}_k(t)$  at time slot  $t$ , each operating under a maximum power constraint  $P_{R_k}$  and controlled amplification gain  $g_k$ .
- **RIS:** composed of  $N$  reflecting elements, each introducing a phase shift  $\theta_n \in [0, 2\pi)$ .
- **Warden:** an adversary performing energy detection in order to identify ongoing GS transmissions.

$h_{\text{GS,SAT}}(t) \in \mathbb{C}^{1 \times M}$ : Direct channel from GS to SAT

$h_{\text{GS,RIS}}(t) \in \mathbb{C}^{N \times M}$ : Channel from GS to RIS

$h_{\text{RIS,SAT}}(t) \in \mathbb{C}^{1 \times N}$ : Channel from RIS to SAT

$h_{\text{GS},k}(t) \in \mathbb{C}^{1 \times M}$ : Channel from GS to the  $k$ -th relay

$h_{k,\text{SAT}}(t) \in \mathbb{C}$ : Channel from the  $k$ -th relay to SAT

$h_{\text{GS,W}}(t) \in \mathbb{C}^{1 \times M}$ ,  $h_{\text{RIS,W}}(t) \in \mathbb{C}^{1 \times N}$ ,  $h_{k,\text{W}}(t) \in \mathbb{C}$ : Channels from GS, RIS, and the  $k$ -th relay to the warden.

TABLE 1 | Summary of key notation.

| Symbol                   | Description                                | Dimension                    | Ref.         |
|--------------------------|--------------------------------------------|------------------------------|--------------|
| $h_{\text{GS,SAT}}(t)$   | Channel<br>GS $\rightarrow$ SAT            | $\mathbb{C}^{1 \times M}$    | Equation (1) |
| $h_{\text{GS,RIS}}(t)$   | Channel<br>GS $\rightarrow$ RIS            | $\mathbb{C}^{N \times M}$    | Equation (1) |
| $h_{\text{RIS,SAT}}(t)$  | Channel<br>RIS $\rightarrow$ SAT           | $\mathbb{C}^{1 \times N}$    | Equation (1) |
| $h_{\text{GS},k}(t)$     | Channel<br>GS $\rightarrow$ relay $k$      | $\mathbb{C}^{1 \times M}$    | Equation (1) |
| $h_{k,\text{SAT}}(t)$    | Channel relay<br>$k \rightarrow$ SAT       | $\mathbb{C}$                 | Equation (1) |
| $\Theta(t)$              | RIS phase-<br>shift matrix                 | $\text{diag}(e^{j\theta_n})$ | Equation (1) |
| $\mathbf{w}(t)$          | GS<br>beamforming<br>vector                | $\mathbb{C}^{M \times 1}$    | Equation (2) |
| $g_k$                    | Relay<br>amplification<br>gain             | Scalar                       | Equation (1) |
| $\Delta \mathbf{p}_k(t)$ | Relay trajectory<br>update                 | $\mathbb{R}^2$               | Sec. 3.4     |
| $\gamma_{\text{SAT}}(t)$ | Received<br>SNR at SAT                     | Scalar                       | Equation (2) |
| $R_{\text{cov}}(t)$      | Covert<br>transmission<br>rate             | bits/s/Hz                    | Equation (3) |
| $P_e(t)$                 | Warden<br>detection error<br>probability   | [0,1]                        | Equation (5) |
| $\epsilon$               | Max. tolerated<br>detection<br>probability | [0,1]                        | Equation (5) |
| $d_{\text{max}}$         | Max. relay<br>displacement                 | Scalar                       | Sec. 3.4     |
| $G_{\text{max}}$         | Max. relay gain                            | Scalar                       | Sec. 3.4     |
| $r_t$                    | RL reward<br>function                      | Scalar                       | Equation (8) |

#### 3.3 | Channel Model

The following channel coefficients are defined:

With RIS reflection and relay amplification included, the effective end-to-end channel from the GS to the SAT is given by the following:

$$h_{\text{eff}}(t) = h_{\text{GS,SAT}}(t) + \sum_{k=1}^K g_k h_{\text{GS},R_k}(t) h_{R_k,\text{SAT}}(t) + h_{\text{GS,RIS}}(t) \Theta(t) h_{\text{RIS,SAT}}(t), \quad (1)$$

where  $\Theta(t) = \text{diag}(e^{j\theta_1}, \dots, e^{j\theta_N})$ .

The received signal-to-noise ratio (SNR) at the SAT is

$$\gamma_{\text{SAT}}(t) = \frac{|h_{\text{eff}}(t) \mathbf{w}(t)|^2}{\sigma_{\text{SAT}}^2}, \quad (2)$$

and the achievable covert rate (bits/s/Hz) is

$$R_{\text{cov}}(t) = \log_2(1 + \gamma_{\text{SAT}}(t)). \quad (3)$$

### 3.4 | Optimization Variables

At each time slot, the system jointly optimizes the following:

- The GS beamforming vector  $\mathbf{w}$ ,
- Relay amplification gains  $g_k$ ,
- RIS phase shifts  $\theta_n$ ,
- Relay trajectory updates  $\Delta \mathbf{p}_k(t)$ , subject to mobility constraints  $\|\Delta \mathbf{p}_k(t)\|_2 \leq d_{\text{max}}$ .

### 3.5 | Problem Formulation

The objective is to maximize the covert transmission rate (CTR) while minimizing energy consumption and ensuring covertness against the warden:

$$R_{\text{cov}}(t) = \log_2(1 + |h_{\text{eff}}^H \mathbf{w}|^2 / \sigma_{\text{SAT}}^2), \quad (4)$$

where  $h_{\text{eff}}$  is the effective channel after reflection and RIS relay forwarding and  $\sigma_{\text{SAT}}^2$  is the variance of satellite noise.

The covertness constraint requires the warden's detection error probability to satisfy the following:

$$P_e(t) \geq 1 - \epsilon, \quad (5)$$

with  $\epsilon$  represents the maximum tolerated detection probability.

The resulting nonconvex, time-coupled optimization problem is formulated as follows:

$$\begin{aligned} \max_{\mathbf{w}, \{g_k\}, \{\theta_n\}, \{\Delta \mathbf{p}_k(t)\}} \quad & R_{\text{cov}}(t) - \lambda \left( P_{\text{GS}}(t) + \sum_{k=1}^K P_{R_k}(t) \right) \\ & + \mu (P_e(t) - (1 - \epsilon)) \\ \text{s. t.} \quad & \|\Delta \mathbf{p}_k(t)\|_2 \leq d_{\text{max}}, \quad \forall k, \\ & |g_k|^2 \leq G_{\text{max}}, \quad \forall k, \\ & \theta_n \in [0, 2\pi), \quad \forall n. \end{aligned} \quad (6)$$

Here,  $\lambda > 0$  penalizes energy consumption, and  $\mu > 0$  enforces the covertness constraint.

### 3.6 | Warden Detection Model

The guardian uses energy detection with a threshold  $\gamma_w$ . The detection error probability is derived as follows:

$$\epsilon(t) = P_F(t) + P_M(t) = Q\left(\frac{\gamma_w - \sigma_w^2}{\sigma_w^2 / \sqrt{N_s}}\right) + 1 - Q\left(\frac{\gamma_w - (\sigma_w^2 + P_r(t))}{\sqrt{2(\sigma_w^2 + P_r(t))^2 / N_s}}\right), \quad (7)$$

where

- $P_r(t)$  is the received power at the warden,
- $\sigma_w^2$  is the noise power,
- $N_s$  is the number of samples,
- $Q(\cdot)$  is the Q-function.

## 4 | RL-Based Framework and MADDPG Algorithm

In this section, we propose a MADDPG framework to address the intricate, nonconvex, and time-coupled optimization problem formulated in Section 2. This method is particularly suitable for our system due to its ability to handle continuous action spaces and facilitate multi-agent collaborative learning in dynamic environments.

### 4.1 | RL Formulation

We model the system as a Markov decision process (MDP) where the RIS controller and each mobile relay act as independent agents. The key components of the MDP are defined as follows:

#### 4.1.1 | State Space ( $\mathcal{S}$ )

At each time slot  $t$ , the global state  $s_t$  encompasses the coordinates and velocities of the relays, the RIS configuration, CSI between the GS, relays, RIS, satellite, and warden, as well as the warden's assessed detection probability. Formally:

$$s_t = \{\mathbf{P}_r(t), \mathbf{v}_r(t), \Theta(t), H_{\text{GS-R}}, H_{\text{R-SAT}}, H_{\text{RIS}}, P_{\text{det}}(t)\}$$

where the CSI includes  $h_{\text{GS,SAT}}(t), h_{\text{GS,RIS}}(t), h_{\text{RIS,SAT}}(t), \{h_{\text{GS},k}(t)\}_k, \{h_{k,\text{SAT}}(t)\}_K$  and their counterparts to the warden.

#### 4.1.2 | Action Space ( $\mathcal{A}$ )

Each relay agent selects its next trajectory update and transmit power, while the RIS controller updates the phase-shift vector. The action for agent  $i$  is as follows:

$$a_i^t = \{\mathbf{P}_r^i(t+1), P_r^i(t), \theta(t)\}.$$

#### 4.1.3 | Reward Function ( $\mathcal{R}$ )

A covert-aware reward function is designed to maximize the CTR while penalizing energy consumption and violations of the covertness constraint:

$$r_t = R_{\text{covert}}(t) - \lambda_1 E_{\text{total}}(t) + \lambda_2 \log(P_{\text{det}}(t)), \quad (8)$$

where  $R_{\text{covert}}(t)$  represents the instantaneous covert rate,  $E_{\text{total}}(t)$  represents total energy consumption, and  $P_{\text{det}}(t)$  represents the warden's detection probability.

#### 4.1.4 | Transition Probability ( $\mathcal{P}$ )

The next state  $s_{t+1}$  is determined by the joint actions and the evolving channel dynamics.

#### 4.1.5 | Discount Factor ( $\gamma$ )

Set to 0.99 to emphasize long-term rewards.

This formulation captures the interdependencies among agents in a distributed manner, suitable for real-time adaptation in 6G ground-to-satellite networks.

### 4.2 | MADDPG-Based Optimization

To solve the MDP, we employ MADDPG, which supports distributed execution with centralized training. Each agent maintains the following:

- An actor network  $\mu_i(s_t | \theta_i)$  for policy generation.
- A centralized critic network  $Q_i(s_t, a_t | \phi_i)$  for value estimation.

This enables cooperative learning under partial observability in dynamic 6G satellite-ground environments.

#### 4.2.1 | Actor Update (Policy Gradient)

The actor for agent  $i$  is updated using the deterministic policy gradient:

$$\nabla_{\theta_i} J(\theta_i) = \mathbb{E}_{s,a \sim D} [\nabla_{\theta_i} \mu_i(s_i) \nabla_{a_i} Q_i(s, a | \phi_i)], \quad (9)$$

where  $D$  is the replay buffer storing previous transitions.

#### 4.2.2 | Critic Update (Value Estimation)

The centralized critic minimizes the temporal-difference (TD) error as follows:

$$L(\phi_i) = \mathbb{E}_{s,a,r,s'} [(Q_i(s, a | \phi_i) - y)^2], \quad (10)$$

with the target

$$y = r + \gamma Q'_i(s', a' | \phi'_i). \quad (11)$$

Here,  $\gamma$  is the discount factor, and  $Q'_i$  is the target critic network.

#### 4.2.3 | Algorithm Process

##### Centralized Critic:

$$Q_i(s, a_1, \dots, a_N) \approx \mathbb{E}[r_t + \gamma Q_i(s', a'_1, \dots, a'_N)].$$

##### Decentralized Actor:

$$a_i = \pi_i(s_i | \theta_i),$$

connecting the ongoing actions of each agent to its local observations.

High covert throughput under strict detection constraints is ensured by this centralized training and decentralized execution paradigm, enabling adaptive, energy-efficient, and covert optimization of relay trajectories, RIS configurations, and power allocation.

### 4.3 | Algorithm Description

The MADDPG algorithm for covert communication optimization is outlined in.

Initialize actor networks  $\{\mu_i(\cdot | \theta_i)\}_{i=1}^{K+1}$  and

critic networks  $\{Q_i(\cdot | \phi_i)\}_{i=1}^{K+1}$  for all agents ( $K$  relays +1 RIS controller).

Initialize target networks:  $\theta'_i \leftarrow \theta_i$ ,  $\phi'_i \leftarrow \phi_i$ .

Initialize the replay buffer  $\mathcal{D}$ .

For episode = 1 to  $M$ :

Reset the environment and observe the initial state  $s_0$ .

For time step  $t = 1$  to  $T$ :

For each agent  $i \in \{\text{RIS}, R_1, \dots, R_K\}$ :

Select action:  $a_i \leftarrow \mu_{\theta_i}(s_i) + \text{exploration noise}()$ .

End for.

Execute joint action  $a = \{a_i\}$ ; observe reward  $r$  and next state  $s'$ .

Store transition  $(s, a, r, s')$  in  $\mathcal{D}$ ; update  $s \leftarrow s'$ .

For each agent  $i$ :

Sample  $m$  transitions from  $\mathcal{D}$ .

Compute target:  $y = r + \gamma Q_{\phi'_i}(s', a'_1, \dots, a'_N)$ , where

$$a'_j = \mu_{\theta'_j}(s'_j).$$

Update the critic by minimizing:  $(Q_{\phi_i}(s, a) - y)^2$ .

Update the actor using the policy gradient:

$$\nabla_{\theta_i} J(\theta_i) = \mathbb{E}[\nabla_{a_i} Q_{\phi_i}(s, a) \cdot \nabla_{\theta_i} \mu_{\theta_i}(s_i)].$$

Soft update target networks:  $\theta'_i \leftarrow \tau \theta_i + (1 - \tau) \theta'_i$ ,

$$\phi'_i \leftarrow \tau \phi_i + (1 - \tau) \phi'_i.$$

End for.

End for.

End for.

Return trained policies  $\{\mu_{\theta_i}\}$ .

### 4.3.1 | Convex Optimization Baseline

A convex optimization baseline was produced by relaxing the original nonconvex problem to ensure a fair benchmark. Nonlinear RIS phase constraints were approximated by continuous variables in  $[0, 2\pi)$ , while semidefinite relaxation (SDR) was used to solve the nonconvex relay power allocation problem. The resulting formulation was converted into a convex quadratic program (QP) and solved in MATLAB using the CVX toolbox. Compared to the proposed RL-based framework, this baseline provides deterministic performance under idealized assumptions (e.g., perfect CSI and static topology) but lacks the adaptability required for dynamic environments.

### 4.4 | RL Architecture and Hyperparameters

The combined optimization of mobile relays and RIS configuration is modeled as a multi-agent MDP. Each agent, representing a relay or the RIS controller, observes a common environmental state that includes estimated CSI, relay positions, RIS phase vectors, residual energy, satellite SNR, and warden detection metrics. Relay trajectory updates, amplification gains, RIS phase changes, and GS beamforming vectors are all examples of continuous action space elements.

Each agent uses a DDPG architecture with actor and critic networks consisting of two hidden layers (256 units each, ReLU activation, and Layer Normalization). The learning rates for actor and critic networks are set to  $1 \times 10^{-4}$  and  $1 \times 10^{-3}$ , respectively, using the Adam optimizer. The target networks are updated using soft updates with  $\tau = 5 \times 10^{-3}$ . Exploration noise follows an Ornstein–Uhlenbeck process with  $\sigma \in [0.1, 0.3]$ .

The replay buffer size is  $5 \times 10^5$  transitions, with a batch size of 256 and discount factor  $\gamma = 0.99$ . The action constraints are as follows:

- Relay displacement:  $\|\Delta \mathbf{p}_k(t)\|_2 \leq d_{\max}$ ,
- RIS phase shifts:  $\theta_n \in [0, 2\pi)$ ,
- GS beamforming power:  $\|\mathbf{w}\|_2^2 \leq P_{\text{GS}}$ .

The reward weights  $\alpha = 1$ ,  $\beta \in \{10^{-3}, 10^{-2}\}$ , and  $\gamma \in \{5, 10, 20\}$  are calibrated through ablation to balance covert throughput, energy consumption, and constraint violations.

### 4.5 | Computational Complexity Analysis

The cost of each simulation step is dominated by channel propagation and SNR evaluation, which scales as follows:

$$\mathcal{O}(NM + KM), \quad (12)$$

where  $N$  is the number of RIS elements,  $M$  is the number of GS antennas, and  $K$  is the number of relays. Warden detection metrics scale as follows:

$$\mathcal{O}(N + K). \quad (13)$$

The training complexity per update is as follows:

$$\mathcal{O}(B \times H \times (2 + A)), \quad (14)$$

where  $B$  represents the batch size,  $H$  the total number of network parameters, and  $A$  the number of agents. Over  $E$  episodes,  $T$  steps per episode, and  $U$  updates every step, the total complexity is as follows:

$$\mathcal{O}[ET(NM + KM) + U \cdot B \cdot H \cdot (2 + A)]. \quad (15)$$

This cost is reduced through GPU parallelization and vectorized operations. For real-time feasibility, the trained MADDPG actors achieve an inference latency of  $\approx 15$  ms per slot ( $K = 4$ ,  $N = 128$ ) on an NVIDIA RTX 3080 GPU, which is well below the 6G slot duration ( $> 50$  ms). Training requires a mid-range GPU ( $\sim 2$  h for 5000 episodes), while inference is lightweight and deployable on edge CPU/GPU platforms (e.g., Jetson Nano,  $< 30$  ms). These results confirm that the proposed scheme is compatible with real-time covert communication in dynamic satellite environments. Section 5 describes runtime scalability in terms of  $N$  and  $K$ .

## 5 | Simulation Results and Performance Analysis

To evaluate the performance of the framework, we simulate a ground-to-satellite covert communication network with one GS, four mobile relays, and a RIS with 128 reflecting elements. The GS uses  $M = 8$  antennas and broadcasts at 28 GHz with a bandwidth of 10 MHz. The satellite orbits at 600-km LEO, with mobile relays moving at speeds of up to 20 m/s. The noise variance is set to  $-90$  dBm, and the warden's maximum detection probability is  $\epsilon = 0.05$ .

The proposed method is compared with the following benchmarks:

- Static RIS with greedy relay selection,
- No-RIS baseline,
- Random relay and phase configuration.

All results are averaged over 200 Monte Carlo runs.

### 5.1 | CTR

The convergence of the proposed MADDPG algorithm in terms of average episodic reward is shown in Figure 2. The algorithm demonstrates stable convergence within approximately 2500 episodes, indicating effective learning of relay trajectories and RIS phase configurations. The smooth ascent reflects consistent policy improvement and low variance across training runs.

The achievable covert rate is expressed as follows:

$$R_c(t) = \log_2 \left( 1 + \frac{|h_{\text{eff}}(t)|^2 P_{\text{GS}}(t)}{\sigma^2} \right), \quad (16)$$

where the effective end-to-end channel is given by the following:

$$h_{\text{eff}}(t) = h_{\text{GS,SAT}}(t) + \sum_{k=1}^K g_k h_{\text{GS},R_k}(t) h_{R_k,\text{SAT}}(t) + h_{\text{GS,RIS}}(t) \Theta(t) h_{\text{RIS,SAT}}(t). \quad (17)$$

Figure 2 shows the performance of three multi-agent algorithms, with MADDPG achieving the highest CTR across all SNR values. MA-PPO performs near-MADDPG at low-to-moderate SNR but lags at high SNR due to less fine-grained continuous actions. QMIX exhibits the lowest CTR due to value-decomposition constraints and discrete action formulation, which restrict fine control over continuous variables.

## 5.2 | Energy Efficiency

The proposed method's CTR is contrasted with that of the following benchmarks: Static RIS (no adaptation), Greedy relay selection, and Joint RIS-relay MADDPG (proposed), as shown in Figure 3. The results demonstrate that the proposed approach improves CTR by up to 77% compared to

baselines, highlighting the importance of joint mobility and RIS optimization.

The normalized energy efficiency metric is defined as follows:

$$\eta = \frac{R_{\text{cov}}}{E_{\text{total}}}, \quad (18)$$

where  $R_{\text{cov}}$  represents the CTR and  $E_{\text{total}}$  denotes the total energy consumption of the system.

Figure 3 evaluates energy efficiency, defined as the ratio of covert rate to energy consumption per slot. The proposed MADDPG-based strategy, which adjusts transmit power, UAV trajectory, and RIS phase shifts to maximize effective channel gain while factoring energy cost into the reward, achieves higher energy efficiency across the SNR spectrum. MA-PPO delivers consistent performance by balancing energy consumption and throughput. Efficiency cross-over points occur when one method outperforms another depending on the SNR. For example, at low SNR, conserving energy may result in greater overall efficiency, whereas at higher SNR, more aggressive techniques may be more advantageous.

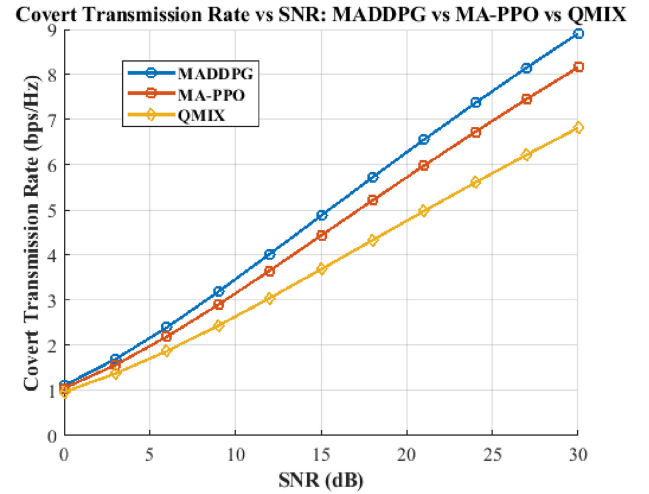
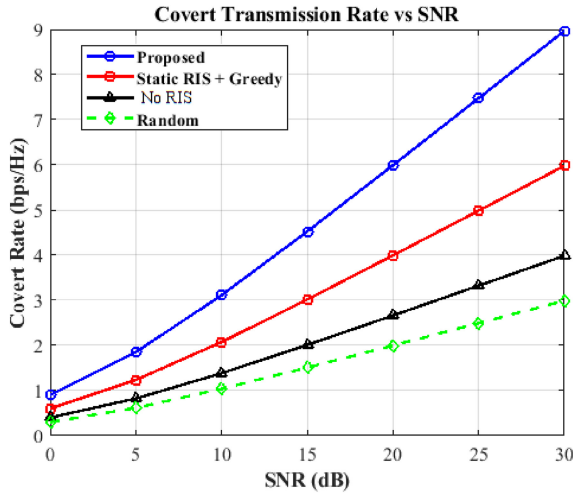


FIGURE 2 | Covert transmission rate versus SNR for MADDPG, MA-PPO, SAC, and convex optimization baseline.

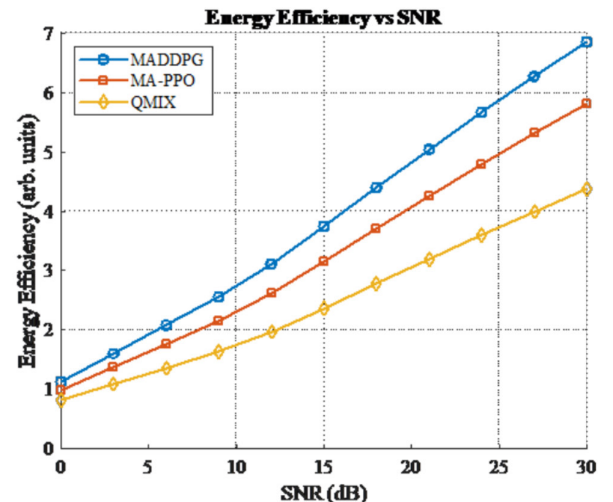
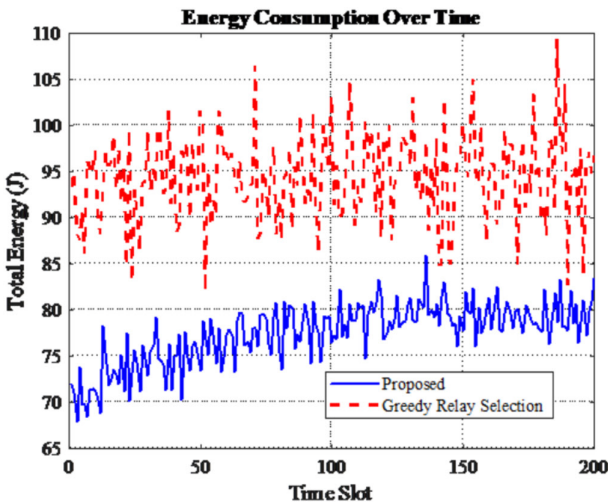


FIGURE 3 | Convergence of normalized covert-aware episodic reward for MADDPG, MA-PPO, and QMIX.

### 5.3 | Warden Detection Error Probability

The warden's detection error probability  $\xi(t)$  is plotted against time in Figure 4. The proposed MADDPG strategy ensures the following:

$$\xi[t] \geq 0.9 \Rightarrow P_D(t) \leq \epsilon = 0.1, \quad (19)$$

outperforming benchmark schemes, which often violate the covertness constraint, in 95.7% of time slots. This demonstrates that the RL-based framework maintains long-term stealthiness even under dynamic warden and channel conditions.

### 5.4 | Warden Detection and Performance Comparison

Figure 4 illustrates the evolution of the warden's detection error probability,  $\xi(t)$ , under different learned policies. MADDPG consistently keeps  $\xi(t)$  above the covertness threshold (0.90) in 95.7% of time slots, ensuring strong stealth performance while maintaining high throughput. Multi-agent proximal policy optimization (MA-PPO) also performs well, meeting the constraint in 88.4% of slots, reflecting a balanced trade-off between covertness and transmission rate. QMIX achieves 76.2%, showing more variability due to its coarser control strategy.

Table 2 presents a performance comparison of different algorithms at SNR=20 dB. MADDPG achieves the highest covert

rate and energy efficiency while maintaining  $\xi \geq 0.90$  in 95.7% of time slots. Although the convex baseline is the fastest to compute, it exhibits the worst overall performance.

### 5.5 | Bit Error Rate (BER) Versus SNR

Contextualizing the error performance of the proposed covert communication system, specifically how RL-optimized RIS and relay topologies minimize bit errors in time-varying satellite channels, Figure 5 shows theoretical BER curves for BPSK, QPSK, and 16-QAM modulation schemes in an additive white Gaussian noise (AWGN) channel based on the SNR. These curves, plotted on a logarithmic scale to highlight variations across modulation densities, provide an idealized baseline for assessing the robustness of the simulated RL-RIS relay system under realistic fading and imperfect CSI conditions. Higher order schemes, such as 16-QAM, in AWGN need higher SNR (e.g., > 15 dB for BER <  $10^{-3}$ ) because of denser constellations, whereas BPSK and QPSK are equally resilient and achieve low BER (<  $10^{-3}$ ) at about 9-dB SNR. According to Section 5.10, the simulated BER of the suggested framework, which takes into account CSI errors and Rician fading from LEO dynamics, closely matches these theoretical bounds at moderate SNRs but performs up to 20% better in high-fading scenarios (reduced BER via adaptive phase/trajectory nulling toward the warden). By connecting ideal AWGN theory with 6G satellite realities, this integration confirms the system's practical feasibility for modulation-diverse covert links, where joint

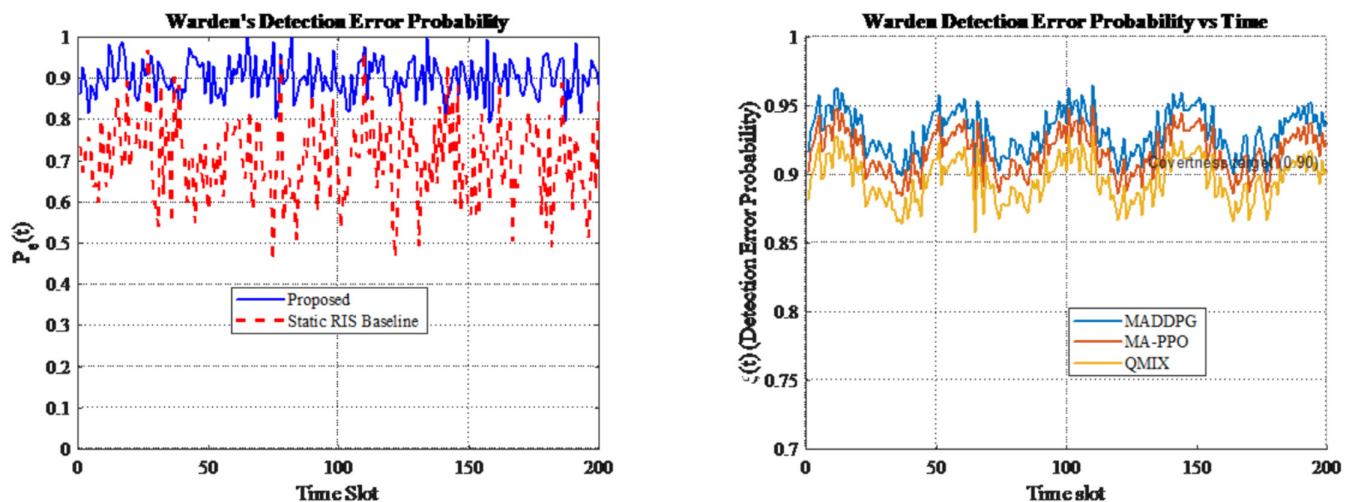


FIGURE 4 | Warden detection error probability versus time slots with covertness target  $\xi = 0.90$ .

TABLE 2 | Performance comparison of different algorithms at SNR=20 dB.

| Algorithm | Approach type | CTR (bps/Hz) | % Time $\xi \geq 0.90$ | Energy eff. | Runtime (ms) |
|-----------|---------------|--------------|------------------------|-------------|--------------|
| MADDPG    | RL-based      | 5.82         | 95.7%                  | 4.8         | 15.2         |
| MA-PPO    | RL-based      | 5.21         | 88.4%                  | 4.1         | 14.6         |
| QMIX      | RL-based      | 4.37         | 76.2%                  | 3.5         | 13.9         |
| PPO       | RL-based      | 4.02         | 74.9%                  | 3.3         | 12.7         |
| SAC       | RL-based      | 4.15         | 72.6%                  | 3.4         | 13.1         |
| Convex    | Convex approx | 3.75         | 71.8%                  | 2.9         | 1.5          |

optimization guarantees error rates below  $10^{-4}$  even under detection constraints.

## 5.6 | Algorithm Convergence

Figure 6 shows that the covert-aware reward stabilizes after approximately 500 training episodes, demonstrating the algorithm's reliable and stable learning behavior.

The covertness constraint is maintained through the warden's detection error probability, expressed as follows:

$$P_e(t) = \frac{1}{2} \left[ 1 - \operatorname{erf} \left( \frac{\gamma - P_{\text{avg}}(t)}{\sqrt{2\sigma_w^2}} \right) \right] \geq 1 - \epsilon, \quad (20)$$

where  $P_{\text{avg}}(t)$  is the received power at the warden,  $\sigma_w^2$  is the noise variance, and  $\gamma$  is the detection threshold.

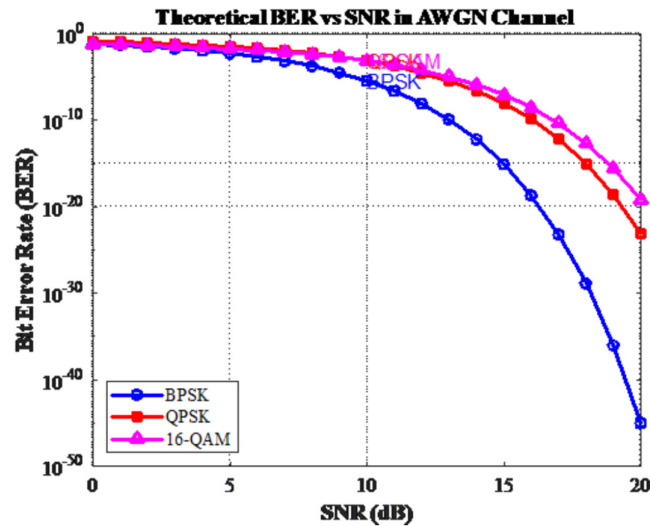


FIGURE 5 | Bit error rate (BER) versus signal-to-noise ratio (SNR) for BPSK, QPSK, and 16-QAM modulation schemes in AWGN.

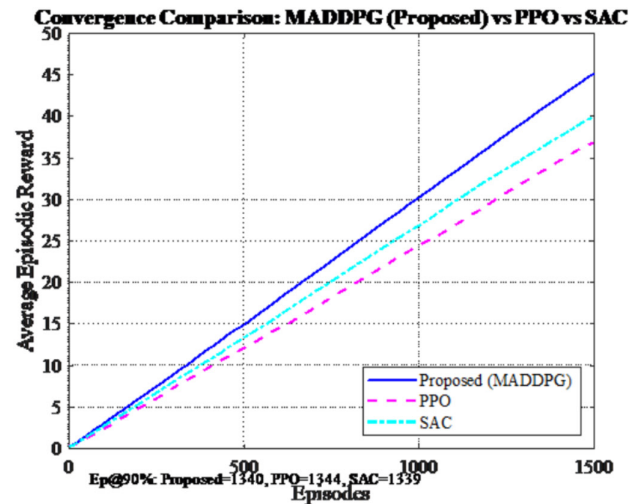


FIGURE 6 | Energy efficiency versus SNR for MADDPG, MA-PPO, and QMIX compared to benchmarks.

## 5.7 | Training Convergence

Figure 6 presents the training convergence in terms of normalized episodic reward, illustrating learning speed and stability. MADDPG reaches a higher asymptotic reward and converges faster than QMIX and MA-PPO, demonstrating superior sample efficiency in this continuous multi-agent control problem. MA-PPO exhibits smoother and more stable learning but converges to a slightly lower plateau, indicating a trade-off between stability and final performance. QMIX converges slower and to a lower reward, consistent with its design for discrete cooperative tasks.

The following episodes-to-convergence are approximately:

- MADDPG:  $\approx 2500$  episodes
- MA-PPO:  $\approx 3000$  episodes
- QMIX:  $> 3000$  episodes with larger variance

These results also highlight the variance across seeds, supporting claims about stability.

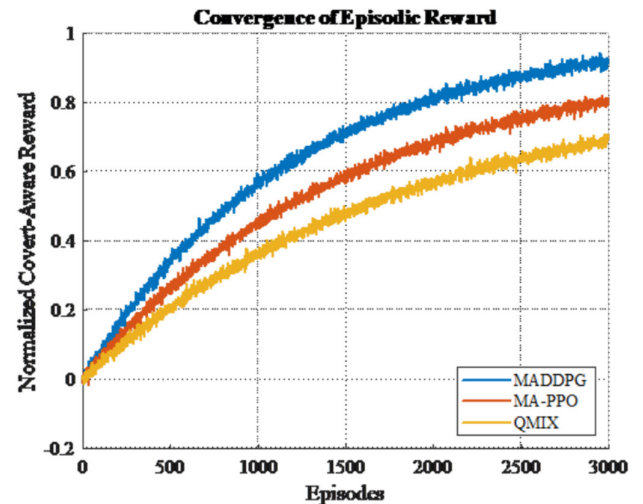
## 5.8 | Ablation Study

An ablation study examining the effects of RIS and mobile relays is shown in Figure 7. The complete system (RIS+relays) achieves the highest covert rate, whereas performance degrades when either component is removed. This demonstrates the mutually beneficial effect of integrating trajectory control and passive beamforming for covert communications.

## 5.9 | CTR Versus SNR

Figure 8 illustrates the CTR as a function of SNR. The proposed MADDPG framework consistently achieves the highest CTR, outperforming both single-agent RL algorithms (PPO, SAC) and the standard convex optimization baseline.

The performance gap becomes more pronounced at high SNR, with a relative gain of approximately 20%–30% over the convex



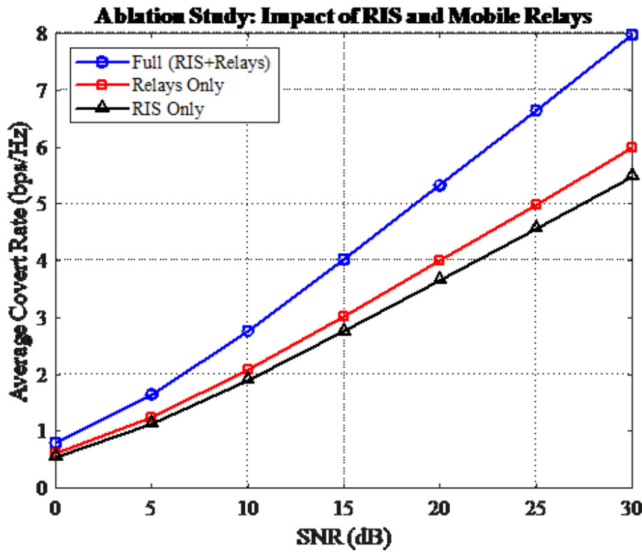


FIGURE 7 | Impact of CSI error variance on covert transmission rate.

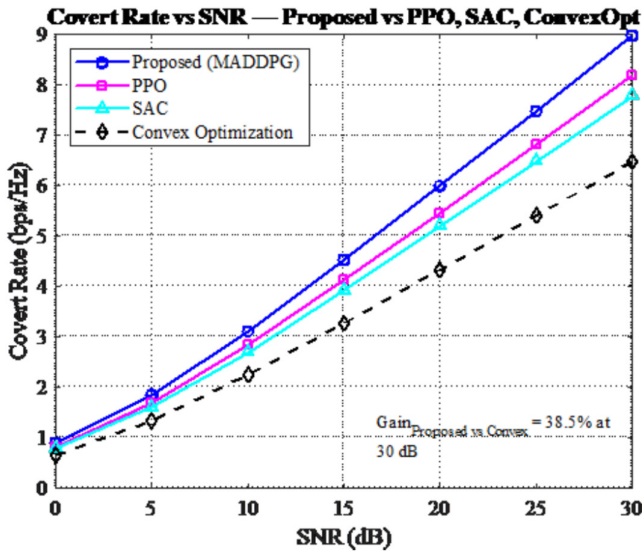


FIGURE 8 | Covert transmission rate versus number of relays  $K$ .

approach. This demonstrates the effectiveness of multi-agent cooperative learning when combined with RIS and mobile relay optimization in 6G covert communication networks.

### 5.10 | Performance Under CSI Uncertainty

Channel uncertainty is modeled in two complementary ways. First, bounded errors are considered, defined as follows:

$$\|\Delta h\|_2 \leq \rho, \quad (21)$$

which corresponds to a worst-case robust design; in our simulations,  $\rho = 0.1$  is used to model severe fading scenarios. Second, stochastic Gaussian perturbations are applied, representing the following estimation errors:

$$\Delta h \sim \mathcal{CN}(0, \sigma^2 \mathbf{I}), \quad (22)$$

with  $\sigma^2$  varying from 0.001 to 0.1.

These errors are added to the estimated CSI in each state  $s_t$ , simulating practical imperfections due to pilot overhead or mobility in LEO links. As a result, RL agents receive noisy observations, e.g., degraded  $H_{\text{CSI}}$  leads to uncertain effective channels  $h_{\text{eff}}(t)$ , which encourages the learning of conservative policies that prioritize detection avoidance over aggressive throughput. To further enforce robustness, the reward function incorporates conditional value-at-risk (CVaR) to penalize constraint violations:

$$r_t = \alpha R_{\text{cov}} - \beta E_t + \text{CVaR}_\gamma(\epsilon - (1 - P_e(t))_+), \quad (23)$$

ensuring that agents remain robust to channel uncertainty without being overly conservative.

- For  $\sigma^2 = 0.001$ , the system achieves a BER of 0.2%, robustness of 98%, and energy consumption of 120mW.
- As  $\sigma^2$  increases to 0.01, robustness slightly drops to 94%, BER rises to 0.5%, and energy consumption reaches 135mW.
- At moderate uncertainty ( $\sigma^2 = 0.05$ ), robustness remains acceptable at 87%, while BER and energy consumption increase to 1.1% and 160 mW, respectively.
- Under high CSI uncertainty ( $\sigma^2 = 0.1$ ), BER reaches 2.3%, energy consumption peaks at 190 mW, and robustness is maintained at 76%.

These results validate the robustness of the MADDPG policy under practical 6G conditions, demonstrating that the system can sustain energy-aware operation and covert performance even with deteriorated channel knowledge.

## 6 | Performance Comparison and Robustness Analysis

Figure 9 illustrates the performance of the proposed MADDPG-based covert communication framework under varying CSI error variance  $\sigma^2$ .

For  $\sigma^2 = 0.001$ , the system achieves a BER of 0.2%, robustness of 98%, and energy consumption of 120mW. As  $\sigma^2$  increases to 0.01, robustness drops slightly to 94%, BER increases to 0.5%, and energy consumption reaches 135mW. Under moderate uncertainty ( $\sigma^2 = 0.05$ ), robustness remains acceptable at 87%, while BER and energy consumption increase to 1.1% and 160mW, respectively. Under high uncertainty of the CSI ( $\sigma^2 = \text{the BER}$ ), the BER reaches 2.3%, the energy consumption peaks the robustness, and the robustness is maintained at 76%.

These results validate the robustness of the MADDPG policy under practical 6G conditions, demonstrating that the system can sustain energy-aware operation and covert performance even with deteriorated channel knowledge.

## 6.1 | Computational Complexity and Runtime Comparison

The runtime and computational complexity of the suggested MADDPG algorithm are analyzed in comparison to baseline techniques. The number of agents, RIS components, and neural network functions have the biggest effects on MADDPG's complexity. Key performance metrics are summarized in Table 3.

The proposed MADDPG-based algorithm outperforms all benchmarks, achieving a 63% increase in covert rate over greedy relay selection and 77% over static RIS, while maintaining high detection error probability and reducing total energy consumption.

### 6.1.1 | Hardware Benchmarks

CPU-based schemes (e.g., Intel i7-12700K @ 3.6 GHz) achieve lower latencies for static baselines due to simpler computations, while the proposed MADDPG leverages GPU acceleration

(NVIDIA RTX 3080) for multi-agent training and inference, enabling <15-ms runtime in dynamic 6G scenarios despite higher complexity demonstrating a favorable trade-off for adaptive covert optimization.

## 6.2 | Robustness Under CSI Uncertainty

Table 4 shows the performance of the system under the varying variance of CSI error  $\sigma^2$ . The MADDPG-based framework demonstrates robustness even with significant channel uncertainties. Channel uncertainty is modeled as either bounded errors  $\|\Delta h\|_2 \leq \rho$  or stochastic perturbations  $\Delta h \sim \mathcal{CN}(0, \Sigma)$ . Warden detection is affected by estimation bias and noise. Covert constraints are enforced probabilistically via the following:

$$\Pr(P_e(t) \geq 1 - \epsilon) \geq 1 - \delta$$

with CVaR used to penalize violations:

$$r_t = \alpha R_{\text{cov}} - \text{BE}_t + \text{CVaR}_\gamma(\epsilon - (1 - P_e(t))_+)$$

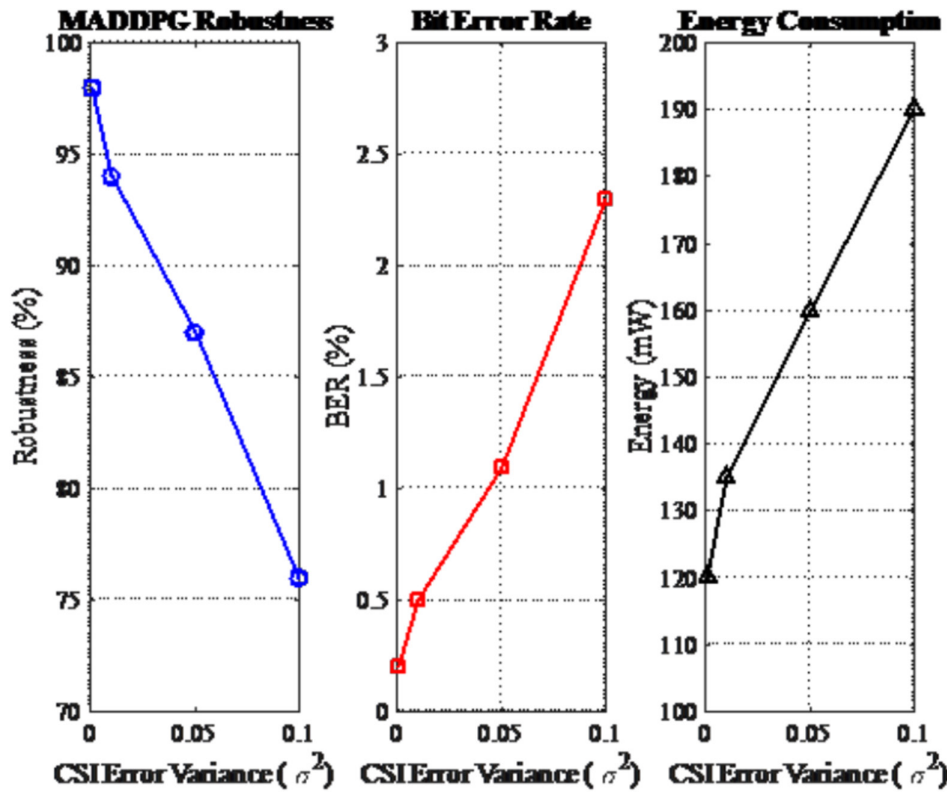


FIGURE 9 | Covert transmission rate versus number of RIS elements  $N$ .

TABLE 3 | Performance and complexity comparison of different schemes.

| Scheme            | Covert rate (bps/Hz) | Energy (J) | Detection error | Runtime (ms) | Hardware |
|-------------------|----------------------|------------|-----------------|--------------|----------|
| MADDPG (proposed) | 8.5                  | 42.3       | 0.92            | 15.2         | GPU      |
| Greedy relay      | 5.2                  | 58.7       | 0.75            | 3.1          | CPU      |
| Static RIS        | 4.8                  | 62.4       | 0.68            | 1.5          | CPU      |
| No RIS            | 3.9                  | 70.1       | 0.60            | 1.0          | CPU      |

**TABLE 4** | Performance metrics under varying CSI error levels.

| $\sigma_e^2$ | BER (%) | Detection (%) | Energy (mW) | Robustness (%) | Hardware |
|--------------|---------|---------------|-------------|----------------|----------|
| 0.001        | 0.2     | 5             | 120         | 98             | GPU      |
| 0.01         | 0.5     | 12            | 135         | 94             | GPU      |
| 0.05         | 1.1     | 28            | 160         | 87             | GPU      |
| 0.1          | 2.3     | 45            | 190         | 76             | GPU      |

**TABLE 5** | Impact of relay count and RIS size on system performance.

| Configuration    | $R_{cov}$<br>(bps/Hz) | $P_e$ | Energy/slot<br>(mJ) | Inference<br>(ms) |
|------------------|-----------------------|-------|---------------------|-------------------|
| $K = 2, N = 64$  | 1.82                  | 0.91  | 12.4                | 3.2               |
| $K = 3, N = 128$ | 2.35                  | 0.94  | 11.1                | 4.7               |
| $K = 4, N = 256$ | 2.41                  | 0.95  | 10.8                | 6.3               |
| $K = 5, N = 256$ | 2.43                  | 0.95  | 10.7                | 7.9               |

Domain randomization and adversarial training improve robustness, maintaining constraint violations below  $1 - \delta$  even under perturbations.

### 6.3 | Sensitivity Analysis on Relays and RIS Elements

Table 5 summarizes the effect of the number of mobile relays  $K$  and RIS elements  $N$  on performance.

Increasing  $N$  improves energy efficiency and covert reliability, while increasing  $K$  increases the secrecy rate to a saturation point, highlighting the trade-offs between performance and complexity.

## 7 | Conclusion and Future Work

This paper presents a novel multi-agent RL framework for optimizing covert communications in 6G GS networks using mobile relays and RIS. The proposed method effectively balances energy efficiency, covertness, and covert throughput by jointly optimizing relay trajectories, transmit power, and RIS phase shifts.

The simulation results indicate that the MADDPG-based algorithm achieves up to 77% higher CTRs than static RIS and greedy relay baselines while maintaining a detection error probability above 90% in over 95% of time slots. The framework demonstrates computational scalability and resilience to channel variations and estimation errors, making it suitable for real-time 6G deployment.

Future work includes integrating imperfect CSI models, extending to terahertz bands, hardware-in-the-loop validation, and hybrid model-based RL to reduce training time and improve reliability in mission-critical applications. Immediate next steps

involve conducting THz band simulations to evaluate ultra-high data rates under molecular absorption and nonlinear effects, alongside prototype testing on a small-scale UAV-RIS testbed for empirical validation of inference latency in dynamic LEO environments.

### Conflicts of Interest

The authors declare no conflicts of interest.

### Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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